

SPG-BANDIT: SKILL PROFILE GUIDED BANDIT FOR TASK SELECTION IN LLM AGENT SKILL EVOLUTION

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1 INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Large language model (LLM) agents have demonstrated competence on complex tasks in diverse environments, including embodied control, web navigation, and software engineering. An important mechanism underlying this breadth of capability is *skill evolution*, through which an agent reflects on execution trajectories, extracts reusable knowledge, and accumulates that knowledge in a skill library for subsequent episodes.

Existing work has primarily addressed two components of this process. The first is *skill quality*, namely the extraction of informative and reusable knowledge from trajectories. For example, SkillRL uses reflection to identify reusable strategies from successful episodes and recurrent failure modes from unsuccessful episodes. The second is *skill utilization*, namely the organization, retrieval, and incorporation of skills into an agent’s inference-time prompt. Representative approaches include structured skill banks with category-specific retrieval, vector-database retrieval, and dynamic prompting.

A critical dimension remains comparatively underexplored: **the order in which tasks are selected for practice**. Existing systems commonly cycle uniformly through a fixed task pool, sample tasks at random, or rely on hand-designed curricula that do not adapt to an agent’s evolving competence. Since tasks can make substantially different contributions to skill progress, an adaptive selection policy may use the interaction budget more efficiently. This motivates the following question: *Can task selection conditioned on an agent’s current skill profile improve the efficiency and effectiveness of skill evolution in LLM agents?*

1.2 CONTRIBUTIONS

We propose **SPG-Bandit** (Skill Profile Guided Bandit), a framework that casts task selection as a contextual bandit problem informed by an agent’s multidimensional skill profile.

Our central idea is to use Multidimensional Item Response Theory (MIRT) to estimate latent proficiency across K skill dimensions and to select tasks that maximize expected progress on dimensions with lower estimated proficiency.

Specifically, this proposal makes the following contributions:

1. **Problem formulation.** We formalize task selection during LLM agent skill evolution as a non-stationary contextual bandit problem. The formulation captures both task-specific differences in expected skill progress and the multidimensional nature of agent competence.
2. **Method.** We propose SPG-Bandit, which combines MIRT-based latent skill profiling with a gap-weighted sliding-window UCB policy. The method maintains an online proficiency estimate and selects tasks with high predicted progress on underdeveloped dimensions while accounting for estimation uncertainty.
3. **General framework.** We provide a modular framework that can be integrated with existing skill-evolution methods and supports systematic evaluation of task-selection efficiency and downstream task performance.

2 PROBLEM FORMULATION

2.1 NOTATION AND SETUP

Consider an LLM agent interacting with a fixed task pool $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_M\}$ containing M tasks. At each round $t = 1, 2, \dots$, the agent selects and executes a task $\tau_t \in \mathcal{T}$ and observes a binary outcome $y_t \in \{0, 1\}$, indicating success or failure given its current skills.

The agent has a *latent skill profile* $\mathbf{s}_t \in [0, 1]^K$, which represents proficiency across K distinct skill dimensions. This profile is unobserved and inferred from task outcomes. We define the estimated *skill progress* after executing τ_t as $\delta_t = \mathbf{s}_{t+1} - \mathbf{s}_t$. Accordingly, δ_t represents the change in the filtered latent-profile estimate following the observed task outcome and reflection update.

Each task τ is associated with a *task embedding* $\mathbf{e}_\tau \in \mathbb{R}^{d_e}$, where $d_e \in \mathbb{N}_+$, typically obtained by applying a sentence encoder to its natural-language goal description.

2.2 PROBLEM DEFINITION

We formalize adaptive task selection as a *non-stationary contextual bandit* problem. At each round t :

1. The agent observes its context: its current skill profile $\mathbf{s}_t$ and the embedding $\mathbf{e}_\tau$ of every task $\tau \in \mathcal{T}$.
2. The agent selects an arm $\tau_t \in \mathcal{T}$.
3. The agent executes τ_t , observes the binary outcome y_t , and updates its skills through reflection.
4. The latent skill profile updates to $\mathbf{s}_{t+1}$.

To prioritize underdeveloped skill dimensions, we define a *gap weight* vector:

$$\mathbf{g}_t = \text{softmax}\left(\frac{\mathbf{1} - \mathbf{s}_t}{\gamma_g}\right), \quad (1)$$

where $\gamma_g > 0$ is a temperature parameter. We use gap-weighted skill progress as the immediate bandit reward:

$$r_t = \mathbf{g}_t^\top \delta_t. \quad (2)$$

The goal is to maximize the expected cumulative reward:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=1}^T r_t \mid \tau_t \sim \pi(\cdot \mid \mathbf{s}_t, \{\mathbf{e}_\tau\}) \right], \quad (3)$$

where π denotes the task-selection policy and T is the number of rounds in the bandit phase. This reward prioritizes progress on underdeveloped dimensions without requiring an auxiliary objective.

We learn a surrogate model $\hat{\delta}_t(\tau)$ and use $\mathbf{g}_t^\top \hat{\delta}_t(\tau)$ as the bandit’s estimated reward.

3 METHOD

3.1 OVERVIEW

SPG-Bandit consists of a warmup phase and a bandit phase. During **warmup**, the agent samples tasks uniformly to obtain an initial set of interaction outcomes. A sequential MIRT model with temporal prior $p(\mathbf{s}_{t+1} \mid \mathbf{s}_t)$ then infers a latent-profile sequence, corresponding skill deltas $\{\delta_t\}$, and task item parameters. These estimates provide supervision for an MLP that maps a task embedding and the current profile to a representation suitable for predicting skill progress.

During the **bandit phase**, the method computes a skill-gap vector $\mathbf{g}_t$ from the current profile and evaluates every candidate task using a sliding-window neural linear UCB score. The selected task maximizes predicted gap-weighted progress together with an uncertainty bonus. The sliding window discards outdated interactions so that the ridge estimator remains responsive to changes in the context-to-progress relationship. After execution, an online Bayesian MIRT update produces the next profile estimate, and the resulting observation is inserted into the window.

3.2 KEY COMPONENTS

Multidimensional IRT (MIRT). We associate each task τ with a K -dimensional discrimination vector $\mathbf{a}_\tau \in \mathbb{R}^K$ and a scalar difficulty $d_\tau \in \mathbb{R}$. Given the agent’s current skill profile $\mathbf{s} \in [0, 1]^K$, its success probability on task τ is:

$$p_\tau(\mathbf{s}) = \sigma(\mathbf{a}_\tau^\top \mathbf{s} - d_\tau), \quad (4)$$

where $\sigma(x) = 1/(1 + e^{-x})$. We impose the element-wise constraint $\mathbf{a}_\tau \geq 0$ and the scalar constraint $d_\tau \geq 0$. The first constraint ensures that increasing any component of $\mathbf{s}$ cannot reduce the predicted success probability. Consequently, the model aligns greater proficiency with stronger task performance. The nonnegative difficulty constraint further encodes a conservative prior over task difficulty and makes $\mathbf{s}$ a directionally interpretable measure of agent capability.

After collecting N warmup outcomes $\{(\tau_t, y_t)\}_{t=1}^N$, we estimate skill profiles and item parameters by alternating optimization. The temporal prior $p(\mathbf{s}_{t+1} \mid \mathbf{s}_t)$ regularizes adjacent profiles and stabilizes inference. As in standard IRT, reliable estimation additionally requires a fixed scale convention and sufficient task coverage.

E-step. With item parameters $\{(\mathbf{a}_\tau, d_\tau)\}$ fixed, infer $\mathbf{s}_{1:N+1}$ by maximizing the joint log-posterior

$$\log p(\mathbf{s}_1) + \sum_{t=1}^N [\log p(y_t \mid \mathbf{s}_t, \mathbf{a}_{\tau_t}, d_{\tau_t}) + \log p(\mathbf{s}_{t+1} \mid \mathbf{s}_t)],$$

where $p(\mathbf{s}_1)$ is a truncated Gaussian prior on $[0, 1]^K$ centered at 0.5. In practice, the temporal prior is also a truncated Gaussian on $[0, 1]^K$, centered at $\mathbf{s}_t$.

M-step. With profiles $\{\mathbf{s}_t\}$ fixed, maximize each task’s observation log-likelihood with respect to $(\mathbf{a}_\tau, d_\tau)$ using L-BFGS-B, subject to $\mathbf{a}_\tau \geq 0$, $d_\tau \geq 0$, and ℓ_2 regularization on $\mathbf{a}_\tau$. Under these constraints, $\mathbf{s} \uparrow \Rightarrow p_\tau(\mathbf{s}) \uparrow$, so greater proficiency monotonically implies a higher predicted success probability.

We alternate the E-step and M-step until the objective converges.

Online update. During the bandit phase, each observation (y_t, τ_t) refines the profile without rerunning the full alternating procedure. Given $\mathbf{s}_t$ and $(\mathbf{a}_{\tau_t}, d_{\tau_t})$, the updated profile is obtained as:

$$\mathbf{s}_{t+1} = \arg \max_{\mathbf{s} \in [0, 1]^K} \log p(y_t \mid \mathbf{s}, \mathbf{a}_{\tau_t}, d_{\tau_t}) + \log p(\mathbf{s} \mid \mathbf{s}_t). \quad (5)$$

Delta prediction via MLP. To predict the improvement associated with a candidate task, we train a two-layer MLP Φ that maps the concatenation of task embedding $\mathbf{e}_\tau$ and current profile $\mathbf{s}$ to a feature vector $\phi \in \mathbb{R}^{d_f}$, where $d_f \in \mathbb{N}_+$. For hidden width d_h , the parameters satisfy $\mathbf{W}_1 \in \mathbb{R}^{d_h \times (d_e + K)}$, $\mathbf{b}_1 \in \mathbb{R}^{d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{d_f \times d_h}$, and $\mathbf{b}_2 \in \mathbb{R}^{d_f}$:

$$\mathbf{h} = \max(\mathbf{W}_1[\mathbf{e}_\tau; \mathbf{s}] + \mathbf{b}_1, 0), \quad (6)$$

$$\phi = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2. \quad (7)$$

A linear pretraining head $\mathbf{W}_3 \in \mathbb{R}^{d_f \times K}$ predicts the expected delta as $\hat{\delta} = \mathbf{W}_3^\top \phi$. The MLP is trained by minimizing mean squared error on the warmup dataset $\{([\mathbf{e}_{\tau_i}; \mathbf{s}_i], \delta_i)\}_{i=1}^N$. After pretraining, $\mathbf{W}_3$ is discarded and Φ is fixed. Bandit decisions use the sliding-window ridge head $\widehat{\mathbf{W}}_t$.

Gap-weighted UCB selection. To accommodate a non-stationary context-to-progress relationship, we maintain a sliding window containing the ℓ most recent *completed* interactions and fit a local ridge-regression head within that window. Let $\mathcal{W}_t$ denote the indices stored in the buffer immediately before bandit round t . The buffer is initialized with the final ℓ warmup observations. For each buffered observation i , define $\phi_i = \Phi([\mathbf{e}_{\tau_i}; \mathbf{s}_i])$. The resulting covariance and response matrices are:

$$\widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t, \quad \mathbf{A}_t = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top, \quad \mathbf{B}_t = \sum_{i \in \mathcal{W}_t} \phi_i \delta_i^\top, \quad (8)$$

For a candidate task τ , let $\phi_{t,\tau} = \Phi([\mathbf{e}_\tau; \mathbf{s}_t])$. The predicted delta is $\hat{\delta}_t(\tau) = \widehat{\mathbf{W}}_t^\top \phi_{t,\tau}$. The matrix $\mathbf{A}_t$ captures epistemic uncertainty through $\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}$. The UCB score is:

$$\text{score}_t(\tau) = \mathbf{g}_t^\top \hat{\delta}_t(\tau) + \beta_t \|\mathbf{g}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}, \quad (9)$$

The first term favors tasks with high predicted gap-weighted progress. The second term is a confidence bonus for the scalarized reward $\mathbf{g}_t^\top \delta_t$, where $\beta_t > 0$ is selected according to a ridge-regression confidence bound.

Sliding-window update. After observing δ_t , we append $(\phi_{t,\tau_t}, \delta_t)$ to the window and evict its oldest entry when the window exceeds size ℓ . The incremental updates are:

$$\mathbf{A}_{t+1} = \mathbf{A}_t + \phi_{t,\tau_t} \phi_{t,\tau_t}^\top - \phi_{\text{old}} \phi_{\text{old}}^\top, \quad (10)$$

$$\mathbf{B}_{t+1} = \mathbf{B}_t + \phi_{t,\tau_t} \delta_t^\top - \phi_{\text{old}} \delta_{\text{old}}^\top, \quad (11)$$

$$\widehat{\mathbf{W}}_{t+1} = \mathbf{A}_{t+1}^{-1} \mathbf{B}_{t+1}. \quad (12)$$

The window size ℓ controls the trade-off between estimation quality, which benefits from a larger sample, and responsiveness to non-stationarity, which benefits from recent observations.

3.3 ALGORITHM

Algorithm 1 summarizes the complete procedure.

4 THEORETICAL ANALYSIS

Because the relationship between context and skill progress may change over time, expected task rewards can be non-stationary. In this section, $t \in \{1, \dots, T\}$ indexes only bandit-phase rounds. We analyze the dynamic *pseudo-regret* of SPG-Bandit under the sliding-window ridge estimator described in Section 3.

4.1 NON-STATIONARY SETTING

For each candidate task τ , define $\phi_{t,\tau} = \Phi([\mathbf{e}_\tau; \mathbf{s}_t])$ and the conditional mean skill progress

$$\boldsymbol{\mu}_t(\tau) = \mathbb{E}[\delta_t(\tau) \mid \mathbf{s}_t].$$

The expected scalar reward is $q_t(\tau) = \mathbf{g}_t^\top \boldsymbol{\mu}_t(\tau)$. Let $\tau_t^* \in \arg \max_{\tau \in \mathcal{T}} q_t(\tau)$ be the dynamic oracle action. We evaluate the dynamic pseudo-regret

$$R_T = \sum_{t=1}^T [q_t(\tau_t^*) - q_t(\tau_t)]. \quad (13)$$

This quantity is a pseudo-regret because it compares conditional expected rewards rather than realized noisy outcomes. Let $\mathcal{F}_{t-1}$ denote the history available before round t . Boundedness of $\mathbf{s}_t$ alone does not imply sublinear non-stationarity. We therefore impose a separate variation condition on the time-varying reward model.

Algorithm 1 SPG-Bandit

Input: Task pool $\mathcal{T}$, warmup steps $N \geq \ell$, bandit horizon T , window size ℓ , feature dimension d_f , gap temperature γ_g , confidence radii $\{\beta_t\}$, regularizer λ , skill dimensions K

Phase 1: Warmup

for $i = 1$ to N **do**

Select $\tau_i \in \mathcal{T}$ uniformly

Execute τ_i , observe $y_i \in \{0, 1\}$, and perform the skill-evolution update

Store (τ_i, y_i)

end for

Run sequential MIRT EM on $\{(\tau_i, y_i)\}_{i=1}^N \rightarrow$ profiles $\{\mathbf{s}_i\}_{i=1}^{N+1}$, deltas $\{\delta_i\}_{i=1}^N$, item parameters $\{\mathbf{a}_\tau, d_\tau\}_{\tau \in \mathcal{T}}$

Train and freeze MLP Φ on $\{([\mathbf{e}_{\tau_i}; \mathbf{s}_i], \delta_i)\}_{i=1}^N$

Initialize $\mathcal{W}$ with the latest ℓ warmup tuples (ϕ_i, δ_i) , where $\phi_i = \Phi([\mathbf{e}_{\tau_i}; \mathbf{s}_i])$

Set $\mathbf{A} = \lambda \mathbf{I}_{d_f} + \sum_{(\phi_i, \delta_i) \in \mathcal{W}} \phi_i \phi_i^\top$, $\mathbf{B} = \sum_{(\phi_i, \delta_i) \in \mathcal{W}} \phi_i \delta_i^\top$, $\widehat{\mathbf{W}} = \mathbf{A}^{-1} \mathbf{B}$, and $\mathbf{s} = \mathbf{s}_{N+1}$

Phase 2: Bandit

for $t = 1$ to T **do**

Compute gap $\mathbf{g}_t = \text{softmax}((1 - \mathbf{s})/\gamma_g)$

for each $\tau \in \mathcal{T}$ **do**

$\phi_{t,\tau} = \Phi([\mathbf{e}_\tau; \mathbf{s}])$, $\hat{\delta}_t(\tau) = \widehat{\mathbf{W}}^\top \phi_{t,\tau}$

$\text{score}_t(\tau) = \mathbf{g}_t^\top \hat{\delta}_t(\tau) + \beta_t \|\mathbf{g}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}^{-1} \phi_{t,\tau}}$

end for

$\tau_t = \arg \max_{\tau \in \mathcal{T}} \text{score}_t(\tau)$

Execute τ_t and observe y_t

$\mathbf{s}_{\text{new}} \leftarrow \text{BAYESUPDATE}(\mathbf{s}, \mathbf{a}_{\tau_t}, d_{\tau_t}, y_t)$

$\delta_t \leftarrow \mathbf{s}_{\text{new}} - \mathbf{s}$ and $\mathbf{s} \leftarrow \mathbf{s}_{\text{new}}$

Append $(\phi_{t,\tau_t}, \delta_t)$ to $\mathcal{W}$

$\mathbf{A} += \phi_{t,\tau_t} \phi_{t,\tau_t}^\top$ and $\mathbf{B} += \phi_{t,\tau_t} \delta_t^\top$

if $|\mathcal{W}| > \ell$ **then**

Pop oldest $(\phi_{\text{old}}, \delta_{\text{old}})$ from $\mathcal{W}$

$\mathbf{A} -= \phi_{\text{old}} \phi_{\text{old}}^\top$ and $\mathbf{B} -= \phi_{\text{old}} \delta_{\text{old}}^\top$

end if

$\widehat{\mathbf{W}} = \mathbf{A}^{-1} \mathbf{B}$

end for

4.2 ASSUMPTIONS

1. **(Time-varying linear delta model.)** There are matrices $\{\mathbf{W}_t^*\}_{t=1}^T$, with $\mathbf{W}_t^* \in \mathbb{R}^{d_f \times K}$, such that $\delta_t(\tau) = \mathbf{W}_t^{*\top} \phi_{t,\tau} + \varepsilon_t(\tau)$, where $\mathbb{E}[\varepsilon_t(\tau) \mid \mathcal{F}_{t-1}, \tau] = \mathbf{0}$.
2. **(Boundedness and noise.)** $\|\phi_{t,\tau}\|_2 \leq 1$, $\|\mathbf{W}_t^*\|_F \leq S$, and, for every $\mathbf{u}$ with $\|\mathbf{u}\|_2 \leq 1$, $\mathbf{u}^\top \varepsilon_t(\tau)$ is conditionally σ -sub-Gaussian.
3. **(Parameter-path variation.)** The reward-model parameters satisfy $P_T = \sum_{t=1}^{T-1} \|\mathbf{W}_{t+1}^* - \mathbf{W}_t^*\|_F$.
4. **(Window coverage.)** For a constant $\kappa > 0$, $\lambda_{\min}(\sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top) \geq \kappa |\mathcal{W}_t|$ whenever $|\mathcal{W}_t| = \ell$.

Assumption 1 matches the algorithmic regression target, which is δ_t rather than $\mathbf{s}_{t+1}$. Assumption 2 controls feature magnitudes, parameter magnitudes, and noise. Assumption 3 quantifies non-stationarity through the parameter path. Assumption 4 prevents an ill-conditioned sliding-window design. For a failure probability $\rho \in (0, 1)$, Appendix A provides a detailed high-probability proof.

4.3 DYNAMIC REGRET DECOMPOSITION

Let $\widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t$. Under Assumptions 1–4, the windowed self-normalized ridge-regression bound in Appendix A gives, with probability at least $1 - \rho$,

$$\left| \mathbf{g}_t^\top (\widehat{\mathbf{W}}_t - \mathbf{W}_t^*)^\top \phi_{t,\tau} \right| \leq \beta_t \|\mathbf{g}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}} + D_t$$

where $\beta_t = \tilde{O}(\sigma \sqrt{K d_f} + \sqrt{\lambda S})$ is a confidence radius and D_t is the bias induced by stale observations. Lemma 1 in Appendix A implies $\sum_{t=1}^T D_t = O(\ell P_T)$. Since the policy selects the task with the largest UCB score, approximate optimism yields

$$q_t(\tau_t^*) - q_t(\tau_t) \leq 2\beta_t \|\mathbf{g}_t\|_2 \sqrt{\phi_{t,\tau_t}^\top \mathbf{A}_t^{-1} \phi_{t,\tau_t}} + 2D_t.$$

Because $\mathbf{g}_t$ is a softmax vector, $\|\mathbf{g}_t\|_2 \leq 1$. Under Assumption 4, the cumulative confidence width is $O(\beta T / \sqrt{\kappa \ell})$, where $\beta = \max_{t \leq T} \beta_t$. Combining the estimation and drift contributions gives:

$$R_T \leq \underbrace{O\left(\frac{\beta T}{\sqrt{\kappa \ell}}\right)}_{\text{estimation}} + \underbrace{O(\ell P_T)}_{\text{drift}}. \quad (14)$$

4.4 OPTIMAL WINDOW AND REGRET BOUND

Balancing the two terms, $\beta T / \sqrt{\kappa \ell} = \ell P_T$, yields the window size:

$$\ell^* = \min \left\{ T, \left(\frac{\beta T}{\sqrt{\kappa} P_T} \right)^{2/3} \right\}, \quad (15)$$

and the corresponding cumulative regret bound:

$$R_T = O \left(\left(\frac{\beta T}{\sqrt{\kappa}} \right)^{2/3} P_T^{1/3} \right). \quad (16)$$

If $P_T = o(T)$ and β, κ^{-1} grow at most polylogarithmically, then the bound is sublinear. For example, if $P_T = O(T^\nu)$ for $\nu < 1$, then

$$R_T = \tilde{O} \left(T^{(2+\nu)/3} \right) = o(T). \quad (17)$$

A DETAILED DYNAMIC PSEUDO-REGRET ANALYSIS

This appendix proves the bound in Section 4. The result is conditional on the time-varying linear delta model and the window-coverage condition stated there. These assumptions are necessary: a changing latent profile alone does not establish a non-stationary bandit regret bound.

A.1 FILTRATION AND CONFIDENCE EVENT

Let $\mathcal{F}_{t-1}$ be the sigma-field generated by all observations and selected tasks before bandit round t . For the selected task, abbreviate $\phi_t = \phi_{t,\tau_t}$. Assumption 1 can be written as

$$\delta_t = \mathbf{W}_t^{*\top} \phi_t + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}, \tau_t] = \mathbf{0}. \quad (18)$$

For $\mathbf{u} \in \mathbb{R}^K$ with $\|\mathbf{u}\|_2 \leq 1$, the scalar noise $\mathbf{u}^\top \varepsilon_t$ is conditionally σ -sub-Gaussian.

Recall the ridge matrices formed from completed observations in $\mathcal{W}_t$:

$$\mathbf{A}_t = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top, \quad \mathbf{B}_t = \sum_{i \in \mathcal{W}_t} \phi_i \delta_i^\top, \quad \widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t. \quad (19)$$

For a failure probability $\rho \in (0, 1)$, define

$$\beta_t = \sigma \sqrt{2K \log \left(\frac{K \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)} + \sqrt{\lambda} S. \quad (20)$$

The factor K follows from applying a scalar self-normalized concentration inequality to each output coordinate and taking a union bound. It can be replaced by a sharper vector-valued confidence radius when additional noise structure is available.

A.2 ESTIMATION ERROR AND DRIFT BIAS

For a candidate τ , let

$$w_t(\tau) = \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}.$$

The following lemma separates stochastic estimation error from bias due to stale samples.

Lemma 1 (windowed prediction error). Under Assumptions 1–4, with probability at least $1 - \rho$, simultaneously for every $t \leq T$ and $\tau \in \mathcal{T}$,

$$\left| \mathbf{g}_t^\top (\widehat{\mathbf{W}}_t - \mathbf{W}_t^*)^\top \phi_{t,\tau} \right| \leq \beta_t \|\mathbf{g}_t\|_2 w_t(\tau) + D_t, \quad (21)$$

where, for a constant $c_{\lambda,\kappa} > 0$ depending only on the regularization and coverage constants,

$$D_t \leq c_{\lambda,\kappa} \sum_{j=\max\{1, t-\ell\}}^{t-1} \|\mathbf{W}_{j+1}^* - \mathbf{W}_j^*\|_F. \quad (22)$$

Proof. Substituting equation 18 into equation 19 and adding and subtracting $\mathbf{W}_t^*$ gives

$$\begin{aligned} \widehat{\mathbf{W}}_t - \mathbf{W}_t^* &= \mathbf{A}_t^{-1} \sum_{i \in \mathcal{W}_t} \phi_i \epsilon_i^\top - \lambda \mathbf{A}_t^{-1} \mathbf{W}_t^* \\ &\quad + \mathbf{A}_t^{-1} \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top (\mathbf{W}_i^* - \mathbf{W}_t^*). \end{aligned} \quad (23)$$

The first two terms are the usual stochastic and regularization terms of ridge regression. Applying the self-normalized inequality coordinatewise, followed by Cauchy–Schwarz over the K output coordinates, bounds their scalarized prediction error by

$$\beta_t \|\mathbf{g}_t\|_2 \|\phi_{t,\tau}\|_{\mathbf{A}_t^{-1}}.$$

For the final term, Assumption 4 bounds the inverse window Gram matrix, while $\|\phi_i\|_2 \leq 1$. Thus, its scalarized prediction error is at most a constant $c_{\lambda,\kappa}$ times

$$\sum_{i \in \mathcal{W}_t} \|\mathbf{W}_i^* - \mathbf{W}_t^*\|_F.$$

By telescoping $\mathbf{W}_i^* - \mathbf{W}_t^*$ along the parameter path, this is bounded by the right-hand side of equation 22. Combining the three bounds proves the lemma. $\square$

Summing equation 22 over time counts each increment $\|\mathbf{W}_{j+1}^* - \mathbf{W}_j^*\|_F$ in at most ℓ windows. Hence,

$$\sum_{t=1}^T D_t \leq c_{\lambda,\kappa} \ell P_T. \quad (24)$$

A.3 FROM CONFIDENCE TO DYNAMIC PSEUDO-REGRET

Define the confidence width

$$U_t(\tau) = \beta_t \|\mathbf{g}_t\|_2 w_t(\tau).$$

On the event in Lemma 1, the expected reward satisfies, for every candidate τ ,

$$q_t(\tau) \leq \mathbf{g}_t^\top \widehat{\mathbf{W}}_t^\top \phi_{t,\tau} + U_t(\tau) + D_t. \quad (25)$$

The selected task maximizes the first two terms on the right-hand side. Applying equation 25 to τ_t^* , then applying the corresponding lower bound from Lemma 1 to τ_t , yields

$$\begin{aligned} q_t(\tau_t^*) - q_t(\tau_t) &\leq \left[\mathbf{g}_t^\top \widehat{\mathbf{W}}_t^\top \boldsymbol{\phi}_{t,\tau_t} + U_t(\tau_t) + D_t \right] - q_t(\tau_t) \\ &\leq 2U_t(\tau_t) + 2D_t. \end{aligned} \quad (26)$$

This is approximate optimism: the stale-sample term D_t prevents exact UCB optimism, but is explicitly controlled by the parameter-path variation.

Because $\mathbf{g}_t$ is a softmax vector, $\|\mathbf{g}_t\|_2 \leq 1$. Once the window is full, Assumption 4 gives $\mathbf{A}_t \succeq \kappa \ell \mathbf{I}_{d_f}$. The warmup initialization supplies a full window before the first bandit decision. Consequently,

$$\sum_{t=1}^T U_t(\tau_t) \leq \frac{\beta T}{\sqrt{\kappa \ell}}, \quad \beta = \max_{t \leq T} \beta_t. \quad (27)$$

Summing equation 26, and using equation 24 and equation 27, proves

$$R_T \leq O\left(\frac{\beta T}{\sqrt{\kappa \ell}}\right) + O(\ell P_T).$$

Balancing the two terms gives the window choice and regret rate in Section 4. $\square$